CORRECTIONS/ERRATA: EST

The lowest signal loss and the highest bandwidth are characteristics of which of the following types of fibers?

A. Air core
B. Multimode
C. Single mode - Ans.
D. Plastic core

Not B.

Which of the following media has the smallest fiber attenuation loss per kilometer?

A. Multimode fiber, 1300 nm wavelength
B. Single mode fiber, 1310 nm wavelength
C. Single mode fiber, 1550 nm wavelength – Ans.
D. Multimode fiber, 850 nm wavelength

Not A.

Fiber optic cables are widely used for high-speed data transport. What is the main role of the glass component in a glass fiber optic cable?

A. Carry the light - Ans.
B. Shock absorber
C. Prevent light leakages
D. Protect the cable itself

Not C.

The antenna beamwidth also known as the half-power beamwidth has control on the antenna's directive gain. If the beamwidth of the antenna decreases what will be the effect to its directive gain?

It will increase. – Ans.

Not decrease..

Correct solution:

Consider a noiseless channel with a bandwidth of 10 kHz is transmitting a signal with four level signal levels, what is the maximum bit rate in kbps the channel can handle?

Solution:

$C = 2BW \log_2 M = 2(10000) \log_2 4$

$C = 40 \text{ kbps} \rightarrow$ Ans.